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Ionic-Liquid-Assisted Preparation of Carbon Nanotube-Supported Uniform Noble Metal Nanoparticles and Their Enhanced Catalytic Performance

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An ionic-liquid-assisted method is developed for the preparation of noble metal nanoparticles supported on multi-walled carbon nanotubes (MWCNTs). The addition of a small amount of ionic liquids to the reaction solution leads to a production of well-crystallized metallic nanoparticles with tunable diameter and narrow size distribution, uniformly dispersed on MWCNTs. The obtained Pt nanoparticles supported on MWCNTs show superior catalytic performance. Both the catalytic activity and stability for the electrochemical oxidation of methanol is remarkably improved. The uniform size of Pt nanoparticles in the composites makes it possible to study the size dependence of the catalytic performance in the selective oxidation of glycerol. The turnover frequency (TOF) has the maximum value at the size of ≈4 nm and the selectivity to glyceraldehyde increases with the particle size.

1. Introduction

Metal nanoparticles supported on carbon nanotubes (CNTs) have shown promising applications in catalysis, electroanalysis, fuel cells, sensors, and surface-enhanced Raman scattering.[1–6] For instance, Pt nanoparticle/CNT composites may be used as the next-generation electrode materials in fuel cells, including direct methanol fuel cells (DMFCs) and proton-exchange membrane fuel cells (PEMFCs).[7,8] CNTs are promising to replace the commercialized carbon black as the catalyst supports for the fuel cells because they can dramatically enhance the utilization, activity, and stability of the catalytic Pt nanoparticles.[7,8] In another field, Pt/CNT composites were recently found to be more active than Pt/amorphous carbon composites in catalyzing glycerol oxidation.[9]

The uniform dispersion of Pt nanoparticles without aggregation on the sidewalls of CNTs and the maintenance of the high electrical conductivity of CNTs are two essential requirements for achieving the best performance in fuel cells.[10] Acid-oxidation has been widely used in the pretreatment of CNTs to introduce binding sites where Pt nanoparticles are anchored. However, this method typically resulted in uneven distribution of surface functional groups as well as severe structural damage to CNTs, hence, is actually inferior for obtaining high catalytic performance. In addition, the catalytic activity and selectivity strongly depend on the size of the metal catalysts.[11] For example, Wang et al. found that Ru/CNT catalyst with a mean Ru particle size of ≈7 nm exhibited the best C_{10}–C_{20} selectivity.[12] Size-controlled preparation of metal nanoparticles with narrow size distribution supported on CNTs is the prerequisite for exploring size-dependent catalytic activity and selectivity of the composites. Therefore, methods that can obtain uniform Pt nanoparticles well dispersed on non-destroyed CNTs are needed.

By grafting polymers such as poly(vinylpyrrolidone) (PVP) on CNTs[10] or modifying CNTs with small organic molecules, such as 1-aminopyrene by π–π stacking,[13,14] uniform decoration of Pt nanoparticles on CNTs have been realized with much less damage to the structure of CNTs. The electrical conductivity of CNTs in resultant composites are well preserved compared with the corresponding acid oxidation-treated CNTs. However, due to the poor conductivity of PVP and 1-aminopyrene molecules, both the conduction between the Pt nanoparticles and the CNTs and that between the neighboring CNTs are obstructed; therefore, the conductivity of the composites may be lowered. This is unfavorable for use as electrocatalysts.

Ionic liquids (ILs) are a new class of organic solvents composed entirely of ions.[15] Because of their high thermal and chemical stability, high ionic conductivity, wide electrochemical windows, and negligible vapor pressure, ILs are expected to be superior solvents for many chemical processes.[15,16] Especially, the fast crystal nucleation rate in ILs normally results in uniform and smaller inorganic nanoparticles with well-crystallized structure.[17] Besides, it is found recently that CNTs can be well dispersed in the imidazolium-based ILs.[18,19] Therefore, introducing ILs into the reaction systems for preparing Pt/CNT composites provides a possible way to obtain composites with excellent catalytic and electrocatalytic performance.[20,21]

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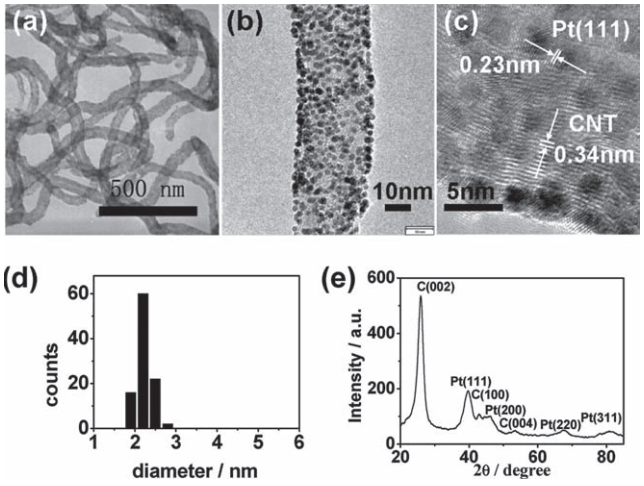

Figure 1. TEM (a) and HRTEM (b,c) images, size distribution histogram of Pt nanoparticles (d), and XRD pattern (e) of the Pt/MWCNT composites obtained by a [BMIM]BF_4-assisited method with 2.0 wt% [BMIM]BF_4 in EG/H_2O (v/v = 2/1) at 115 °C. The Pt loading is ≈20 wt% in the composites.

Here, we report an IL-assisted method for in situ one-pot preparation of metal nanoparticles supported on multi-walled CNTs (MWCNTs). The purified MWCNTs were used directly without acid oxidation or other modification, thus their electrical conductivities are maintained. The metal nanoparticles prepared via this method exhibit monodispersed size distribution and uniform dispersion on MWCNT supports. This one-pot method is not only facile and scalable, but also adapts to prepare various noble metal nanoparticles, such as Pt, Rh, Ru, Ir, and PtRu nanoparticles, supported on MWCNTs. Besides, the size and amount of metal nanoparticles can be easily tuned by the amount of IL, the solvent composition, the reaction temperature, and the concentration of metal precursors. The as-prepared Pt/MWCNT composites show enhanced electrocatalytic activity and stability towards methanol oxidation as well as improved catalytic activity for glycerol oxidation compared with the composites prepared without ILs.

2. Results and Discussion

Figure 1a shows a typical TEM image of Pt/MWCNT composites obtained with the aid of 1-butyl-3-methyl imidazolium tetrafluoroborate ([BMIM]BF_4). The composites are uniformly dispersed without aggregation. Almost all of the nanoparticles are attached on MWCNTs with few dissociative particles. High-resolution TEM image (Figure 1b) of the product clearly shows the highly dispersed nanoparticles with uniform size on MWCNTs. The size distribution of Pt nanoparticles on MWCNTs is obtained by measuring diameters of 100 particles on MWCNTs. All following data in this paper about the size distributions of produced nanoparticles are obtained in a similar way. The diameters of Pt nanoparticles supported on MWCNTs distribute in a narrow size range between 1.8 and 3.0 nm (Figure 1d). The average diameter is 2.21 ± 0.21 nm with a small dispersion coefficient of 0.093. The X-ray diffraction (XRD) pattern of the product reveals the clear diffraction peaks of both carbon nanotubes and Pt nanoparticles. Moreover, from the high-resolution transmission electron microscopy (HRTEM) image of the composites (Figure 1c), one can clearly see the lattice of carbon nanotubes of 0.34 nm, which is equal to the interlayer distance of the MWCNT walls. The wall structure of MWCNTs remains intact.

As shown in Figure S1a in the Supporting Information, Pt/AO-MWCNT composites, that is, Pt nanoparticles supported on acid-oxidized MWCNTs, prepared by similar method except without IL additives are poorly dispersed and typically aggregated to large regiments. The aggregation may reduce the active surface area of nanoparticles. Moreover, the diameters of Pt nanoparticles on MWCNTs are obviously less uniform compared with those in the composites obtained by [BMIM]BF_4-assisited method (Figure S1b and Figure 1b). Their sizes are between 2 and 6 nm (Figure S1c) with a mean size of 3.62 ± 0.81 nm and a dispersion coefficient of 0.22, which is much larger than the coefficient of Pt nanoparticles supported on MWCNTs synthesized with [BMIM]BF_4. As reported previously, if the MWCNTs were used without acid-oxidation pretreatment, the prepared Pt/MWCNT composites were in worse dispersion and more Pt nanoparticles aggregated.[22]

To clarify the effect of [BMIM]BF_4 in the synthetic procedures of Pt/MWCNT composites, several control experiments have been conducted. In the first control experiment, we used other ILs such as [BMIM]PF_6, [HMIM]PF_6, or [N-BuPy]BF_4 instead of [BMIM]BF_4 to assist the Pt nanoparticle formation on MWCNTs. The three kinds of IL all have imidazolium or pyridine cations, which may interact with CNTs.[19] As shown in Figure S2a, b, and c, the composites obtained by IL-assisted procedures are all in good dispersion. Also, nanoparticles are highly-dispersed on MWCNTs without aggregation and the sizes of particles are all within a narrow distribution. The diameters of Pt nanoparticles in the three composites are 1.99 ± 0.26 nm, 2.36 ± 0.24 nm, and 2.78 ± 0.29 nm and the corresponding dispersion coefficients are 0.13, 0.10, 0.10, respectively. Therefore, these three ILs exhibit similar functions as [BMIM]BF_4 during Pt nanoparticle formation on MWCNTs. Then, $NaBF_4$, which has the same anion as [BMIM]BF_4, was added instead of [BMIM]BF_4 in the preparation procedures of Pt/MWCNT composites. In contrast to the composites prepared from the IL-assisted processes, Pt/MWCNT composites made from this route are seriously aggregated with most nanoparticles agglomerate together (Figure S2d). The sizes of Pt nanoparticles also distribute in wide range (2.0–6.0 nm) with a mean value of 3.62 ± 0.87 nm and a dispersion coefficient of 0.24 (Figure S2e,f). Such composites are similar to the composites produced without any additives (Figure S1). This experiment reveals that BF_4^- anions do not contribute to the improved dispersion of the composites and the mono-dispersed size of Pt nanoparticles on MWCNTs. From these two control experiments, it is concluded that IL cations play key roles in obtaining well-dispersed uniform Pt nanoparticles supported on MWCNTs.

UV–vis absorption spectra of a series of solutions were measured to investigate the interaction between IL cations and K_2PtCl_6. The results are shown in Figure S3. The $PtCl_6^{2-}$ ions in EG/H_2O (v/v = 2/1) solution display two absorption bands at 208 nm and 263 nm, respectively, which is consistent with those

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reported in the literature.[23] After adding [BMIM]BF_4, a new absorption peak appears at 222 nm, which is ascribed to the absorption peak of $[BMIM]^+$ ions. We then added MWCNTs into K_2PtCl_6 solution. After sonication for 15 min and centrifugation, the upper solution exhibits an absorption curve nearly the same as that of $PtCl_6^{2-}$ ions. If MWCNTs are added into the solution of mixed K_2PtCl_6 and [BMIM]BF_4, the upper solution after centrifugation shows the absorption curve nearly the same as that of K_2PtCl_6 and [BMIM]BF_4 solution. However, if the solution containing MWCNTs, K_2PtCl_6 and [BMIM]BF_4 is sonicated for 15 min, the upper solution after centrifugation shows quite different absorption curve. The absorption intensity of peak at 263 nm is found to drop ≈53% compared with that of the original solution. As IL cations such as $[BMIM]^+$ can be adsorbed on MWCNTs,[19] it is possible that $PtCl_6^{2-}$ ions are attracted to the surface of MWCNTs by the electrostatic interaction between $[BMIM]^+$ and $PtCl_6^{2-}$. Thus, the concentration of $PtCl_6^{2-}$ ions in the upper solution after centrifugation decreases a lot, which results in the drop of the peak intensity at 263 nm. Besides, the sonication process facilitates the dispersion of the MWCNTs in the solution as well as the adsorption of $[BMIM]^+$ and $PtCl_6^{2-}$ onto the surface of MWCNTs.

Control experiments without MWCNTs were also conducted. The nanoparticles obtained without IL aggregate into congeries as large as hundreds of nanometers, while most nanoparticles are well-separated when IL is added (Figure S4). This indicates that IL is able to adsorb onto the surface of Pt nanoparticles to prevent them from aggregating.[24]

In addition, we studied the effect of IL amount on the dispersion of NPs on MWCNTs and the size distribution (**Figure 2** and Figure S5). When 0.5 wt% of IL was added into the solution, the diameter of the NPs decorated onto MWCNTs decreases from 3.62 ± 0.81 nm to 2.56 ± 0.41 nm. Also, the corresponding dispersion coefficient decreases from 0.22 to 0.16. Increasing the IL amount further does not show any additional influence on the morphology of the composites (Figure 2e). This result shows that an IL amount of as small as 0.5 wt% is enough to get the ideal Pt/MWCNT composites. This will provide a greener and cheaper method for producing Pt/MWCNT composites than the procedures performed in the pure IL solvents.[25]

From the above control experiments, it can be concluded that ILs play three roles in the preparation of Pt/MWCNT composites. First, the ILs can help the dispersion of MWCNTs, which also facilitates the uniform loading of the Pt nanocrystals on CNTs. Second, the effective adsorption of IL cations, such as $[BMIM]^+$, $[HMIM]^+$ and $[N\text{-}BuPy]^+$, on MWCNTs greatly increases the binding sites for metal precursors.[19] Thus, $PtCl_6^{2-}$ ions can be uniformly adsorbed onto the surface of MWCNTs via the electrostatic interaction between $[BMIM]^+$ and $PtCl_6^{2-}$. Third, IL molecules can adsorb on the surface of formed Pt nanoparticles to act as protecting agents,[24] which will effectively eliminate the aggregation of nanoparticles.

The size of Pt nanoparticles supported on MWCNTs can be easily tuned by the solvent composition, reaction temperature, and the amount of Pt. The nucleation and growth in EG solvent without H_2O is expected to be much more difficult than that in EG/H_2O (v/v = 2/1) solvent. When only EG is used as the solvent, the mean diameter of Pt nanoparticles in the composites decreases to 1.64 ± 0.43 nm with 99% of the particles in the range from 0.7 to 2.5 nm (Figure S6a,b). However, if the preparation is conducted in EG at 160 °C instead of at 120 °C, the average diameter of Pt nanoparticles in the composites increased to 4.23 ± 0.71 nm with 99% in the range 3.0–6.0 nm (Figure S6c,d). Moreover, decreasing the Pt loading amount by using a low concentration of Pt precursor results in smaller Pt nanocrystals on MWCNTs. For instance, the mean diameter changes from 2.49 ± 0.35 nm to 2.10 ± 0.33 nm when the Pt loading amount is changed from 30% to 2.0% (Figure S6e, and f).

Importantly, this one-pot IL-assisted method is not only facile and scalable, but also adapts to prepare various noble metal nanoparticles or alloy nanoparticles supported on MWCNTs. For example, Ru nanoparticles with a size of 3.11 ± 0.49 nm, Rh nanoparticles with a size of 3.51 ± 0.42 nm, Ir nanoparticles with a size of 1.80 ± 0.27 nm and PtRu nanoparticles with a size of 2.10 ± 0.42 nm, can all be prepared on MWCNTs by [BMIM]BF_4-assisited method (**Figure 3**, Figure S7 and Figure S8). The nanoparticles in all of the composites show monodispersed size distribution and uniform dispersion on MWCNTs. When $RuCl_3$, $RhCl_3$ or $IrCl_3$ is used as the metal precursor, it is

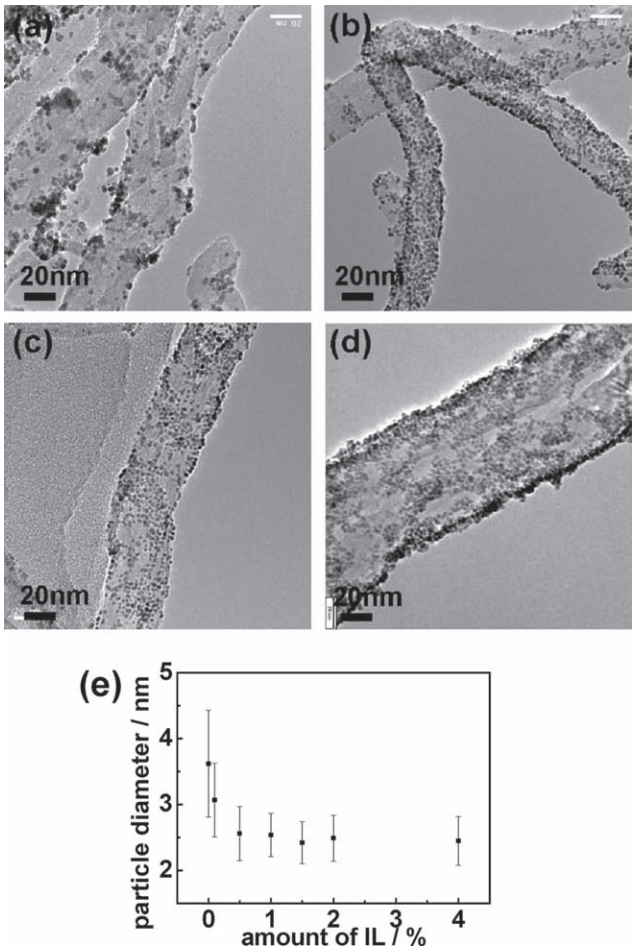

Figure 2. TEM images of the Pt/MWCNT composites obtained by [BMIM]BF_4-assisited method in EG/H_2O (v/v = 2/1) with different amount of [BMIM]BF_4 at 120 °C. The amount of [BMIM]BF_4 used is 0.1 wt% (a), 0.5 wt% (b), 1.0 wt% (c), and 4.0 wt% (d), respectively. Pt loading is ≈30 wt% in the composites. e) The [BMIM]BF_4 amount dependence of the Pt nanoparticle diameter.

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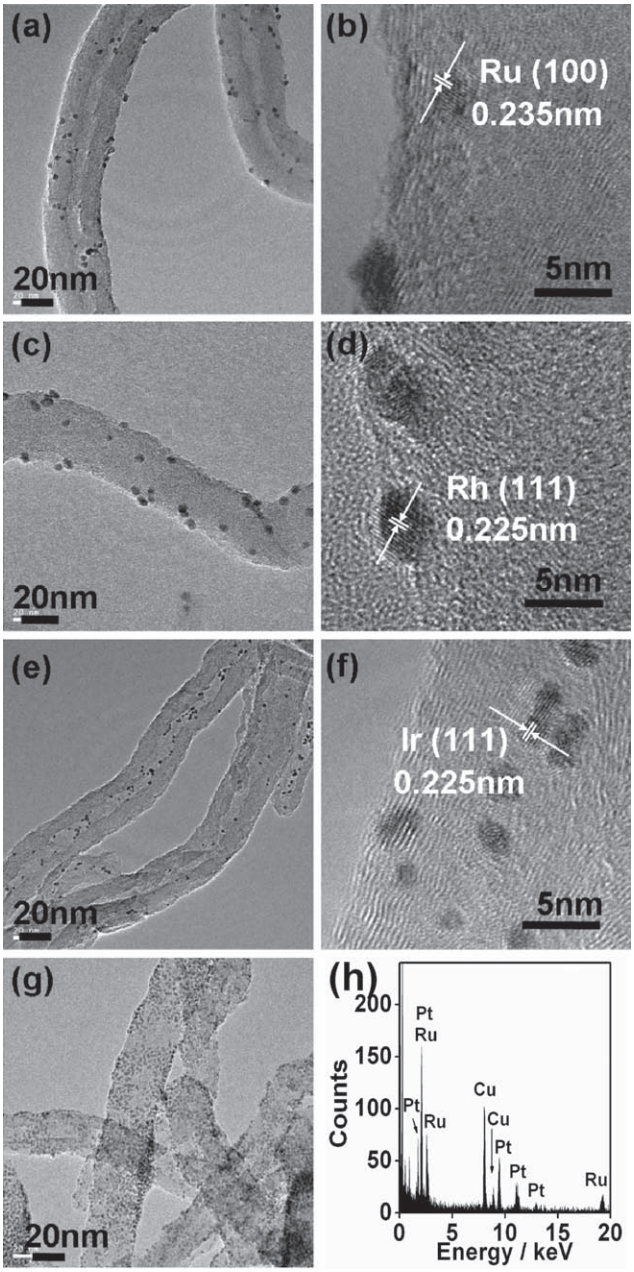

Figure 3. TEM images and HRTEM images of Ru (a,b), Rh (c,d), Ir (e,f) and PtRu (g) supported on MWCNTs. h) is EDX spectrum of PtRu/MWCNT composites. The copper signal came from the copper grid.

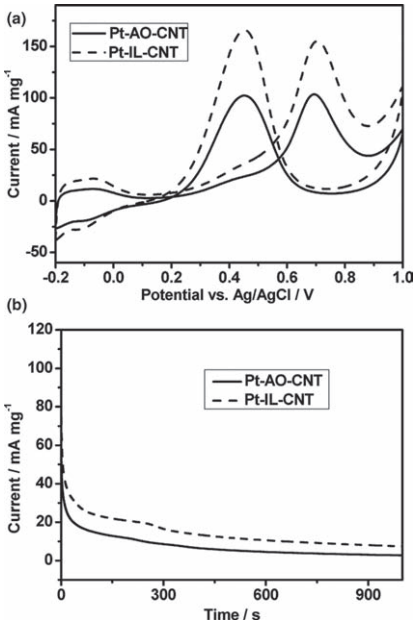

Figure 4. Cyclic voltammograms (a) and current-time curves (b) of the Pt/MWCNT composites obtained by the [BMIM]BF_4-assisited method (red dashed curves) and the acid-oxidation-assisted method (black solid curves) in solution of 0.5 M H_2SO_4 and 1.0 M CH_3OH.

necessary to add HCl into the reaction system to form $RuCl_4^-$, $RhCl_4^-$ or $IrCl_4^-$ ions, which can adsorb on the surface of MWCNTs via the electrostatic interaction with $[BMIM]^+$. Otherwise, Ru, Rh or Ir nanoparticles can hardly grow on MWCNTs. It further confirms the importance of the electrostatic interaction between the IL cations and metal precursor anions.

The electrocatalytic performance of the Pt/MWCNT composites in the methanol oxidation reaction was evaluated by cyclic voltammetry. As shown in **Figure 4**a, the forward peak current value of the composites prepared by [BMIM]BF_4-assisited method is 155 mA mg^{-1} of Pt, about 50% higher than 103 mA mg^{-1} of Pt for the composites yielded via acid-oxidation-assisted method. The enhancement of the electro-catalytic activity of the Pt/MWCNT composites prepared by [BMIM]BF_4-assisited method may arise from the following two aspects. First, the well-dispersed Pt nanoparticles on the surface of MWCNTs render more accessible surfaces of Pt nanoparticles for the methanol oxidation reaction, accordingly leading to their higher electrocatalytic activity. Second, the [BMIM]BF_4-assisited method requires no pretreatment of MWCNTs with acid, which may effectively keep the high conductivity of the MWCNTs, Therefore, the fast charge transfer through the composites may results in their high electrocatalytic activity. Third, the small size of Pt nanoparticles in the composites may also contribute to the improvement of the electrocatalytic performance of the composites toward methanol oxidation reaction.[10] Figure 4b compares current–time curves of the two Pt/MWCNT composites toward methanol oxidation at 0.68 V versus Ag/AgCl reference electrodes. Compared with the composites obtained by acid-oxidation-assisted method, the current value of the Pt/MWCNT composites prepared by [BMIM]BF_4-assisted method attenuates at a much slower rate. The steady current values at 1000 s for the composites prepared by [BMIM]BF_4-assisted method and acid-oxidation-assisted method are 7.43 mA mg^{-1} and 2.76 mA mg^{-1}, respectively, showing the remarkably improved electrocatalytic stability of the Pt/MWCNT composites prepared in the presence of ILs.

The monodispersed Pt nanoparticles with tunable sizes on MWCNTs also facilitate the study of their size-dependent catalytic activity and selectivity, as an example, for glycerol oxidation. It is known that glycerol is a potent bio-building block and its oxidation can lead to the sustainable synthesis of many fine chemicals and intermediates currently derived from fossil

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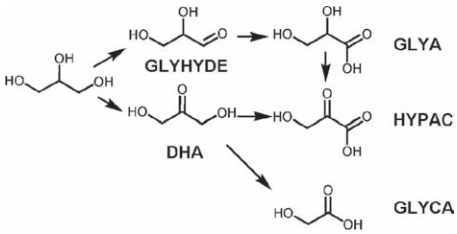

Figure 5. Scheme of the reaction pathways of glycerol oxidation.

resources. Due to the complexity and multiple pathways of this reaction (**Figure 5**), a formidable challenge remains to achieve high selectivities to the target products for further applications.[26] Oxidation of the primary hydroxyl groups (1-OH) of glycerol forms glyceraldehyde, and its further oxidation product of glyceric acid, while dihydroxyacetone is obtained from oxidation of the secondary hydroxyl groups (2-OH) of glycerol. Recent studies have revealed that Pt nanoparticles is active and selective for glycerol oxidation.[9,11,27] However, size effect of the Pt particles and their activity and selectivity is unclear because the Pt nanoparticles employed in the literature typically possess wide size distributions on the supports.[9,11]

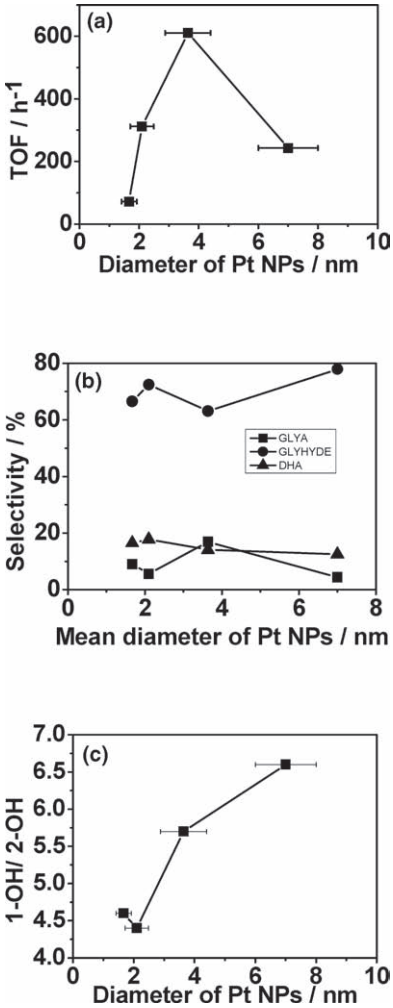

Figure 6. Effect of Pt particle diameter on TOF (a), selectivity towards main products (b) and the ratio between 1-OH oxidation products and 2-OH oxidation products (c). The ratio in (c) was calculated as: (selectivity of GLYA + selectivity of GLYHYDE)/(selectivity of DHA).

The size effect is clearly observed here on the Pt/MWCNT composites with well-controlled Pt nanoparticle size (1.7–7.0 nm), prepared by the [BMIM]BF_4-assisited method. As shown in **Figure 6**a, the activity normalized by exposed surface Pt atoms (estimated by using the mean diameters of the Pt particles and by assuming that they are spherical), i.e., turnover frequency (TOF), increases from 243 to 611 h^{-1} with decreasing the particle size from 7.0 to 3.6 nm. The TOF value declines rapidly to 312 and 72 as the particle size decreases further to 2.1 and 1.7 nm. Such size effect on the activity has been found prevalently with nanoparticle catalysts in many reactions, such as low-temperature CO oxidation on Au,[28] low-temperature aqueous phase Fisher–Tropsch synthesis on Ru,[29] and aerobic oxidation of alcohols on Pd.[30]

Concerning the selectivity, glycerol oxidation normally forms glyceric acid as the major product on Pt.[9,11] Differently, glyceraldehyde, a thermodynamically less stable product relative to glyceric acid, is obtained preferentially on the Pt nanoparticles supported on MWCNTs, as shown in Figure 6b, as we found in our recent communication.[27] Other products mainly include dihydroxyacetone and glyceric acid. The selectivities to these products also depend sensitively on the size of the Pt particles in the range 1.7–7.0 nm (Figure 6b). The selectivity to glyceraldehyde change inversely with that to glyceric acid as the particle size varies. The highest selectivities to glyceraldehdye and glyceric acid are obtained at 7.0 and 3.6 nm, resepectively. The highest glyceric acid selectivity for the catalyst with 3.6 nm Pt particles is consistent with its highest activity, leading to favorable oxidation of glyceraldehyde to glyceric acid. The selectivity to dihydroxyacetone decreases with increasing the particle size, reflecting the steric hindrance of the oxidation of the 2-OH relative to the 1-OH in glycerol. These results indicate that smaller Pt particles favor the oxidation of the secondary hydroxyl groups. Such effect can be seen more clearly from the selectivity ratios of 1-OH oxidation products (glyceraldehyde and glyceric acid) to dihydroxyacetone, which increase with the particle size (Figure 6c). Such size effect on both the activity and selectivity provides directions for controlled synthesis of Pt catalysts more efficient glycerol oxidation.

3. Conclusions

An IL-assisted method has been developed for the preparation of highly dispersed and uniform noble metal nanoparticles with tunable size supported on MWCNTs. The Pt/MWCNT composites prepared by this IL-assisted method not only show enhanced electrocatalytic activity and stability toward methanol oxidation, but also exhibit remarkably size-dependent catalytic activity and selectivity for glycerol oxidation. It is expected that this kind of noble metal nanoparticle/CNT composites will find applications in fuel cells and industrial catalysis.

4. Experimental Section

Materials: MWCNTs were purchased from Shenzhen Nanotech. Port. Co., Ltd (China) and used as received if not pointed out. 1-butyl-3-methyl

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imidazolium tetrafluoroborate ([BMIM]BF_4), 1-butyl-3-methyl imidazolium hexafluorophosphate ([BMIM]PF_6), 1-hexyl-3-methyl imidazolium hexafluorophosphate ([HMIM]PF_6), and *N*-butyl-pyridine tetrafluoroborate ([N-BuPy]BF_4) were purchased from Henan Lihua Pharmaceutical Co. Ltd. (P. R. China) and used as received. Other reactants and solvents are of analytical grade and used without further purification.

Preparation of Pt/MWCNT Composites and PtRu/MWCNT Composites by an IL-assisted Method: In a typical process, 20 mg MWCNTs were added to the mixed solvent of 20 mL ethylene glycol (EG) and 10 mL water. The suspension was then sonicated for ≈5 min after 0.65 g ionic liquid (2.0 wt% of the solution, [BMIM]BF_4 was used typically) was added. The suspension was sonicated for another ≈3 min after 10 mg or 15 mg K_2PtCl_6 was added. Then the reaction mixtures were kept at a proper temperature in oil bath for 4 h. The black precipitates were separated by centrifugation and washed several times by water, ethanol and acetone, and then dried at 80 °C to get the products. Similar procedures were used for the preparation of PtRu/MWCNT composites.

Preparation of Ru/MWCNT Composites, Rh/MWCNT Composites and Ir/MWCNT Composites by an[BMIM]BF4-Assisited Method: 2 wt% [BMIM]BF_4 was added to the suspension containing 10 mg MWCNTs and 20 mL EG. After the suspension was sonicated for 5 min, 1.5 mg $RuCl_3$ and 0.5 mL 0.10 M HCl were added into the suspension and was sonicated for another 3 min. The mixture was refluxed at 195 °C in oil bath for 3 h, after which the black precipitates were separated by centrifugation and washed several times by water, ethanol and acetone, then dried at 80 °C to get the Ru/MWCNT composite products. Similar procedures were used for the preparation of Rh/MWCNT composites and Ir/MWCNT composites using $RhCl_3$ and $IrCl_3$ as precursors. The reflux temperatures were 197 °C and 185 °C, respectively.

Preparation of AO-MWCNTs:[22] 10 mg MWCNTs were added to ≈18 mL mixed-acid solution containing 8.0 M HNO_3 and 8.0 M H_2SO_4. The suspension was sonicated in a water bath with a temperature of 60 °C for 2 h. Then, the black precipitates were separated by centrifugation and washed several times by water, then dried to get AO-MWCNTs.

Characterization of the Composites: The obtained products were characterized with XRD (Rigaku Dmax-2000 with Cu Kα radiation, λ = 1.5406 Å), TEM (JEOL 200 CX), HRTEM (JEOL H9000 and Tecnai F30), ICP (Leeman Profile) and absorption spectrometry (Perkin Elmer Lambda 35).

Electrochemical Measurements: 10 µL suspension containing 25 µg Pt/MWCNT composites was transferred onto a glassy carbon electrode with a diameter of 3.0 mm. The electrode was then dried and served as working electrode, and immersed in 50 mL solution of 0.5 M H_2SO_4 and 1.0 M CH_3OH. An Ag/AgCl electrode was used as the reference electrode and a Pt foil was employed as the counter electrode. Electrochemical measurements were conducted with an electrochemical workstation (CHI660C). Measurements were performed at a scan rate of 50 mV s^{-1} at room temperature under N_2 atmosphere.

Catalytic-Property Measurements: Oxidation of glycerol was carried out at atmospheric pressure in a thermostated flask (100 cm^3) equipped with a magnetic stirrer, thermocouple, and gas-supply system. ≈40 cm^3 of 2 wt% glycerol aqueous solution and 0.025 g of catalyst were added into the reactor. An constant flow of oxygen (65 cm^3 min^{-1}) was introduced into the solution when the temperature reached 90 °C. After the reaction, the aqueous solution was separated by filtration and analyzed using a high-performance liquid chromatography with photodiode array detector (PAD) and Alltech OA-1000 column (HPLC, Shimadzu LC-20A series). All products were confirmed with UV–vis absorption spectra and retention times of the standard products purchased from Alfa, Sigma, and TCI.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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